

Microfinance Loan Repayment Prediction Project

Problem Statement: Predict whether a customer will repay a loan within 5 days.

Dataset: Synthetic dataset with loan amount, income, age, tenure, previous loans, repayment history.

Models Trained: Logistic Regression, Random Forest, Gradient Boosting, SVM.

Evaluation Metrics: ROC-AUC and LogLoss for model comparison.

Results:

LogisticRegression: ROC_AUC=0.923, LogLoss=0.357

RandomForest: ROC_AUC=1.000, LogLoss=0.023

GradientBoosting: ROC_AUC=1.000, LogLoss=0.001

SVM: ROC_AUC=0.998, LogLoss=0.065

Conclusion: The model can be used to predict loan repayment probabilities and assist MFIs in better customer selection.